FACULTY
OF MATHEMATICS
AND PHYSICS
Charles University

MASTER THESIS

Bc. Kamil Belán

Numerical methods for the simulation of air flow past a mountain

Mathematical Institute

Supervisor of the Thesis doc. RNDr. Michal Pavelka, Ph.D.
Study Programme Mathematical and Computational Modelling in Physics

Prague 2025
(Draft - April 14, 2026)

Declaration of Authorship

I declare that I carried out this master thesis on my own, and only with the cited sources, literature and other professional sources. I understand that my work relates to the rights and obligations under the Act No. 121/2000 Sb., the Copyright Act, as amended, in particular the fact that the Charles University has the right to conclude a license agreement on the use of this work as a school work pursuant to Section 60 subsection 1 of the Copyright Act.

Place & Date *Bc. Kamil Belán*

Write dedication here.

Bc. Kamil Belán

Information

Title	Numerical methods for the simulation of air flow past a mountain
Author	Bc. Kamil Belán
Institute	Mathematical Institute
Supervisor	doc. RNDr. Michal Pavelka, Ph.D. — Mathematical Institute
Keywords	Manual, Demo, Draft, WIP

Abstract

Write abstract here.

Contents

Information & Abstract **iv**

Contents **v**

Introduction **vi**

1 Flows around a topography **1**

1.1 Physical description 1

Observed phenomena 1 • Governing equations 3 • Mountain waves 3

1.2 Mathematical modelling of atmospheric flows . 4

Initial and boundary conditions 4 • Discretization 4 • Well-balancing 4

2 SPH **5**

2.1 Overview of SPH 5

Pros and cons 5 • Uses 5

2.2 Cornerstones of SPH 5

Partition 5 • Kernel interpolation 5 • Differential operators 5

2.3 SPH formulations 5

WCSPH 5 • Hopkins 5

2.4 Additional techniques 7

Packing algorithm 7 • Adaptive smoothing length 7 • Initial configuration 7

3 Experiments **8**

4 Voronoi **9**

Summary and Outlook **10**

References **11**

Introduction

The goal of this thesis is to investigate the use of certain numerical methods in simulating atmospheric flows around topography, the so-called *orographic flows*. The text is structured as follows:

(1) chapter Chapter 1 →p.1 provides an introduction to the physics of *orographic flows*;
(2) chapter Chapter 2 →p.5 summarizes the main numerical method used, that is the *Smoothed Particles (SPH)*;
(3) chapter Chapter 4 →p.9 introduces a different numerical method: *Semi-Implicit Lagrangian Voronoi Approximation (SILVA)*;
(4) chapter Chapter 3 →p.8 presents these numerical methods through numerical experiments and discusses the obtained results.

Motivation

Earth's atmosphere is one of the most thoroughly studied and well-described natural systems. With the development of computer architecture, a great effort has been made to complete the *experimental* data with the *predictive* abilities of computational modelling. The need for effective and reliable predictive power is naturally well-founded – many processes in the atmosphere have a massive influence on the lives of Earth's inhabitants.

One example of such a process is orographic flow – an atmospheric flow of air mass over topography. Such flows are fairly common in nature and most importantly they give rise to effects of high relevance in atmospheric dynamics & weather forecasting, such as

(1) downslope windstorms;
(2) atmospheric internal gravity waves[1];
(3) clear air turbulence.

The ability to obtain accurate and reliable *numerical* solutions to such flows is thus highly *practically relevant.* [2] In this work, we aim to explore the suitability of SPH and SILVA for this task.

[1] Note the distinction between *gravitational waves* – disturbances of *space-time* studied in the general theory of relativity, and *gravity waves* – disturbances of a *fluid* studied mainly in meteorology and oceanology. To make this distinction even more evident, the name *internal* gravity waves is sometimes used in the literature.

[2] And scientifically relevant also: the study of orographic flows and mainly internal gravity waves is an active field in the atmospheric science community.

1 Flows around a topography

The first chapter is devoted to the specification and description of the studied problem. Its structure is as follows

(1) important phenomena associated with orographic flows are described in Section 1.1 →p.1;
(2) equations governing the flow are formulated in 4 →p.3;
(3) a more detailed description of mountain waves is carried out in 4 →p.3;
(4) section Section 1.2 →p.4 discusses suitable boundary conditions the system needs to be accompanied by;
(5) section Section 1.2 →p.4 is devoted to the *well-balanced* property of certain discretisation schemes.

TODO: SPLIT MATHS AND PHYSICS?

1.1 Physical description

Let us begin by describing orographic flows from the point of physics, providing not only further motivation, but only important physical aspects of the studied problem.

Observed phenomena

Internal gravity waves

Internal gravity wave: is a type of wave phenomenon arising under certain (although common) atmospheric conditions [3]. Orographic flows typically induce two types of internal gravity waves: *mountain waves* and *lee waves* p. 60 [1]. Although IGWs generated by a flow over terrain influence mainly the dynamics of atmosphere locally near that terrain, their contribution *cannot be neglected* also in global climate and weather forecasts, due to the energy transfer across layers of the atmosphere that is typical for gravity waves p. 58 [1].

[3] More detail on these conditions is given in the section Chapter 1 →p.1.

Downslope windstorms

Under specific physical conditions, air carried by a flow over a mountain obstacle can arrive warmer and drier at the lee side of the mountain, manifesting as *warm* gusts of air descending

Figure 1.1: Lenticular clouds formed under the influence of mountain waves.

Source: [2]

from mountains into valleys. These type of winds are called **(1)** chinook in North America; **(2)** föhn in Europe. The interference of such winds with mountain and lee waves can manifest as gusts of hurricane-level force p. 307 [3], posing a great threat to the inhabitants and infrastructure of nearby regions.

TODO: FIGURES

Clear-air turbulence

As mentioned, IGWs serve as an important mechanism of energy transfer across layers of the atmosphere. If particular conditions are met, it can happen that the wave propagating upwards grows in amplitude. Breakdown of these waves can produce a specific kind of turbulence, the so called *clear-air turbulance*. This turbulence is extremely dangerous to aerial transport, as its presence is not accompanied by any kind of visual effects, hence the name p. 58 [1]. A model that accurately captures mountain waves *could* be helpful in predicting areas of CAT.

TODO: FIGURES

Governing equations

Mechanics

$$\mathbb{T} = -p\mathbb{I} + 2\mu\mathbb{D},$$

where $p = p(\rho, \theta)$ is the thermodynamic pressure, $\mathbb{I}$ is the identity matrix, μ is the dynamic viscosity and $\mathbb{D} = \operatorname{sym} \nabla \mathbf{v}$ is the symmetric part of the velocity gradient.

$$\partial_t(\rho\mathbf{v}) + \nabla \cdot (\rho\mathbf{v} \otimes \mathbf{v}) = \nabla \cdot \mathbb{T} + \rho\mathbf{b},$$

with $\mathbf{b}$ being the specific density of the body forces, *e.g.* gravity.

$$\partial_t\rho + \nabla \cdot (\rho\mathbf{v}) = 0.$$

Thermodynamics

We have two equations of state:

$$p(\rho, \theta) = R\rho\theta,$$

$$e(\rho, \theta) = zR\theta.$$

Hydrostatic balance

Mountain waves

When an air parcel hits a mountain barrier, the topography can act as a ramp and eject the air parcel into higher altitude. In the case of a *stably stratified atmosphere*, the buoyancy force decreases with altitude, and thus the net force acting on the air parcel points downwards. On its way to lower altitude, the air parcel overshoots its equilibrium position, which in turn leads to the *increase* of buoyancy force, resulting in a total acceleration pointing upwards. This buoyancy-driven oscillatory motion can be thought of as a prototype of an *internal gravity wave*. When the initial displacement from the equilibrium state of the air parcel is caused by a flow over topography, the resulting gravity wave is called *a mountain wave*. p. 143 [3] A different type of gravity waves associated with orographic flows are *lee waves*: gravity waves trapped between the ground and upper levels of the atmosphere due to reflection p. 60 [1].

Stratification

Potential temperature

1.2 Mathematical modelling of atmospheric flows

Initial and boundary conditions

Discretization

Well-balancing

2 SPH

2.1 Overview of SPH

Pros and cons

Uses

2.2 Cornerstones of SPH

Partition

Kernel interpolation

Differential operators

2.3 SPH formulations

WCSPH

Hopkins

Pressure - entropy

Recall that the entropy of an isentropic flow of an ideal gas has the form[4]

$$\eta = \eta_0 + c_v \log\left(\frac{P}{\rho^\gamma}\right)$$

and $\eta =$ const . We see that the quantity A defined as [4]

$$A = \frac{P}{\rho^\gamma},$$

is in fact equal to

$$A = \exp\left(\frac{\eta - \eta_0}{c_v}\right),$$

meaning that the flow is *isentropic*, if and only if the quantity A is *conserved* under the flow. Thus A is an *entropy-like* variable: conserved under isentropic flows.

[4] This is a test side note.

The choice

$$x_i = m_i A_i^{\frac{1}{\gamma}}, \tag{2.1}$$

gives the pressure P_i as

$$P_i = \left(\sum_{j=1}^{N} m_j A_j^{\frac{1}{\gamma}} w_{ij}(h_i) \right)^{\gamma} \tag{2.2}$$

and the evolution equations as

$$\frac{\mathrm{d}\mathbf{v}_i}{\mathrm{d}t} = -\sum_{j=1}^{N} m_j (A_i A_j)^{\frac{1}{\gamma}} \left(\frac{f_i P_i}{P_i^{\frac{2}{\gamma}}} \nabla_i w_{ij}(h_i) + \frac{f_j P_j}{P_j^{\frac{2}{\gamma}}} \nabla_i w_{ij}(h_j) \right), \tag{2.3}$$

with

$$f_i = \left(1 + \frac{h_i}{2n_i} \frac{\partial n_i}{\partial h_i} \right)^{-1}$$

Pressure – potential temperature

Recall the *potential temperature* θ

$$\theta = T \left(\frac{P_0}{P} \right)^{\frac{R}{c_p}},$$

where T is the gas's thermodynamic temperature, and P_0 is the reference pressure. [5] Using the thermal state equation

$$P = R\rho T,$$

together with the fact

$$\frac{R}{c_p} = \frac{\gamma - 1}{\gamma},$$

we obtain

$$\theta = \frac{P}{\rho R} P_0^{\frac{\gamma-1}{\gamma}} P^{\frac{1-\gamma}{\gamma}} = \frac{P_0^{\frac{\gamma-1}{\gamma}}}{R} \left(\frac{P^{\frac{1}{\gamma}}}{\rho} \right) = \frac{P_0^{\frac{\gamma-1}{\gamma}}}{R} \left(\frac{P}{\rho^{\gamma}} \right)^{\frac{1}{\gamma}} = \frac{P_0^{\frac{\gamma-1}{\gamma}}}{R} A^{\frac{1}{\gamma}},$$

[5] Potential temperature is the thermodynamic temperature the air parcel would possess if *adiabatically* moved from the thermodynamic state (T, P) to the state (θ, P_0).

so setting

$$\tilde{\theta}_0 := \frac{P_0^{\frac{\gamma-1}{\gamma}}}{R}$$

we obtain

$$\theta = \tilde{\theta}_0 A^{\frac{1}{\gamma}}. \tag{2.4}$$

This implies the potential temperature θ is also *an entropy-like* variable - it is *conserved* under isentropic flows.

Rewriting 2.1 using 2.4 we obtain

$$x_i = \frac{m_i}{\tilde{\theta}_0}\,\theta_i. \tag{2.5}$$

Using this quantity the pressure 2.2 can then be expressed as

$$P_i = \left(\sum_{j=1}^{N} \frac{m_j}{\tilde{\theta}_0}\,\theta_j w_{ij}(h_i)\right)^{\gamma}. \tag{2.6}$$

From 2.3 we obtain

$$\frac{\mathrm{d}\mathbf{v}_i}{\mathrm{d}t} = -\sum_{j=1}^{N} m_j\,\frac{\theta_i\theta_j}{\tilde{\theta}_0^2}\left(\frac{f_i P_i}{P_i^{\frac{2}{\gamma}}}\,\nabla_i w_{ij}(h_i) + \frac{f_j P_j}{P_j^{\frac{2}{\gamma}}}\,\nabla_i w_{ij}(h_j)\right), \tag{2.7}$$

with **??** unchanged

$$f_i = \left(1 + \frac{h_i}{2n_i}\,\frac{\partial n_i}{\partial h_i}\right)^{-1}. \tag{2.8}$$

2.4 Additional techniques

Packing algorithm

Adaptive smoothing length

Initial configuration

3 Experiments

4 Voronoi

Summary and Outlook

Summary and Outlook.

References

Back-references to the pages where the publication was cited are given by •.

[1] C. J. Nappo. An Introduction to Atmospheric Gravity Waves. 2nd ed. Elsevier, **2013**. 359 pp. 1, 2, 3

[2] joiseyshowaa. Lenticular Clouds in Iceland. **2018**.
URL:
https://www.flickr.com/photos/joiseyshowaa/43381939781/
(visited on 04/14/2026) 2

[3] Bruce Sutherland. Internal Gravity Waves. Includes bibliographical references and index
Stratified fluids and waves – Interfacial waves – Internal waves in uniformly stratified fluid – Nonlinear considerations – Generation mechanisms – Wave propagation and spectra The study of internal gravity waves provides many challenges: they move along interfaces as well as in fully three-dimensional space, at relatively fast temporal and small spatial scales, making them difficult to observe and resolve in weather and climate models. Solving the equations describing their evolution poses various mathematical challenges associated with singular boundary value problems and large amplitude dynamics. This book provides the first comprehensive treatment of the theory for small and large amplitude internal gravity waves. Over 120 schematics, numerical simulations and laboratory images illustrate the theory and mathematical techniques, and 130 exercises enable the reader to apply their understanding of the theory. This is an invaluable single resource for academic researchers and graduate students studying the motion of waves within the atmosphere and ocean, and also mathematicians, physicists and engineers interested in the properties of propagating, growing and breaking waves. Cambridge University Press, **2010**. 1 p. 2, 3

[4] Philip F. Hopkins. A General Class of Lagrangian Smoothed Particle Hydrodynamics Methods and Implications for Fluid

Mixing Problems. *Monthly Notices of the Royal Astronomical Society*, **2013**. Comment: 16 pages, 16 figures, MNRAS. Matches accepted version (tests added, typo fixed). A version of the public GADGET-2 modified to use the 'pressure-entropy' formulation proposed here can be downloaded at http://ascl.net/1305.006.
DOI: 10.1093/mnras/sts210 ARXIV: 1206.5006 [astro-ph]
(visited on 01/08/2026) 5